

Progress Test 2.A — Elec-S1

2.A · Characteristic curve and experimental evidence for Ohm’s law
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Reading a slope (1 mark)

The characteristic $i = f(u)$ of a resistor is a straight line through the origin, of slope 20 mA V^{-1} . Give its resistance R .

[Empty grid for Exercise 1]

Marking

①
 0 1
 [] []

Exercise 2. Characteristic of a component — definition (5 marks)

Define the characteristic of a two-terminal component. Which convention is used to plot the characteristic of a passive component? Draw it and give the sign of the power received. What is the shape of the characteristic of an ohmic conductor?

[Empty grid for Exercise 2]

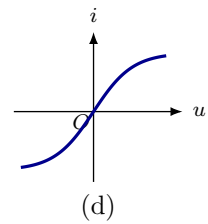
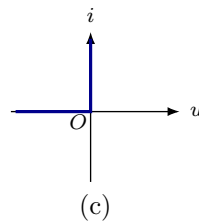
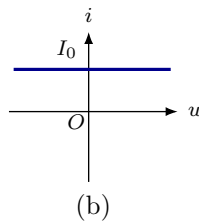
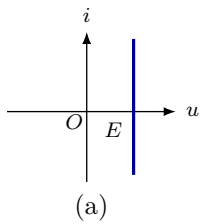
Marking

①
 0 1 2
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 ②
 0 1 2
 [] [] []
 ③
 0 1
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Exercise 3. Identifying characteristics

(3 marks)

The four characteristics below are plotted in the (u, i) plane.



Characteristic (a) is that of:

- ☐ an ohmic conductor
 ☐ an ideal voltage source
☐ an ideal current source
 ☐ a lamp

Characteristic (c) is that of:

- ☐ a Zener diode ☐ a lamp
☐ an ideal diode ☐ an ideal current source

Which statement is correct?

- ☐ Every characteristic is a straight line through the origin.
- ☐ A straight-line characteristic through the origin is the signature of an ohmic conductor.
- ☐ The characteristic of a lamp is a straight line through the origin.
- ☐ A characteristic does not depend on how u and i are oriented.

Exercise 4. Estimating a resistance from measurements

(5 marks)

The characteristic of a resistor was measured in the receiver convention. The measurements at low current are the noisiest.

i_k (mA)	1.0	2.0	4.0
u_k (V)	0.70	0.90	2.00

(a) Compute the naive estimate $R_{\text{naive}} = \frac{1}{N} \sum_k u_k / i_k$. (b) Compute the least-squares estimate $\hat{R} = \sum_k u_k i_k / \sum_k i_k^2$. (c) Which value should be kept? Justify.

[illegible]

Marking

1

01

2

01

3

01

4

01

5

01

— End of the test —